

Assignment 1

ADVANCED DIFFERENTIAL EQUATIONS AND MODELLING

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S5221080

Question 1

From the information Moira has provided, the following scaled competition model has been developed to describe the population dynamics of drop bears, k , and bunyips, b ,

$$\frac{dk}{dt} = k(3 - k - b)$$

$$\frac{db}{dt} = b(4 - 2b - 2k)$$

This model can be used to explore the population dynamics and ultimately, provide Moira with advice.

- (a) To determine the equilibrium points for this non-linear system of differential equations that models the population dynamics for species that compete for the same resource, each equation is set to zero, i.e., $\frac{dk}{dt} = 0$ and $\frac{db}{dt} = 0$

$$\text{Let } \frac{dk}{dt} = 0$$

$$k(3 - k - b) = 0$$

$$\therefore k = 0 \quad \text{or} \quad 3 - k - b = 0$$

$$k + b = 3$$

$$k = 3 - b$$

$$\text{Let } \frac{db}{dt} = 0$$

$$b(4 - 2b - 2k) = 0$$

Firstly, if $k = 0$,

$$b(4 - 2b - 2(0)) = 0$$

$$b(4 - 2b) = 0$$

$$\therefore b = 0 \quad \text{or} \quad 4 - 2b = 0$$

$$2b = 4$$

$$b = 2$$

$\therefore (k, b) = (0, 0)$ and $(0, 2)$ are equilibrium points

Next, if $k = 3 - b$,

$$b(4 - 2b - 2(3 - b)) = 0$$

$$b(4 - 2b - 6 + 2b) = 0$$

$$b(-2) = 0$$

$$-2b = 0$$

$$b = 0$$

To find the corresponding k value, substitute $b = 0$ back into the expression for k

$$k = 3 - 0$$

$$k = 3$$

$\therefore (k, b) = (3, 0)$ is an equilibrium point

Therefore, there are three equilibrium points: $(0, 0)$, $(0, 2)$, $(3, 0)$.

- (b) Finding the eigenvalues and eigenvectors of the corresponding linear system for each equilibrium point.

First find the Jacobian matrix of the system, where

$$J(k, b) = \begin{pmatrix} \frac{\partial f}{\partial k}(k, b) & \frac{\partial f}{\partial b}(k, b) \\ \frac{\partial g}{\partial k}(k, b) & \frac{\partial g}{\partial b}(k, b) \end{pmatrix} \quad \begin{aligned} f &= k(3 - k - b) = 3k - k^2 - kb \\ g &= b(4 - 2b - 2k) = 4b - 2b^2 - 2kb \end{aligned}$$

$$J(k, b) = \begin{pmatrix} \frac{\partial f}{\partial k}(3k - k^2 - kb) & \frac{\partial f}{\partial b}(3k - k^2 - kb) \\ \frac{\partial g}{\partial k}(4b - 2b^2 - 2kb) & \frac{\partial g}{\partial b}(4b - 2b^2 - 2kb) \end{pmatrix}$$

$$J(k, b) = \begin{pmatrix} 3 - 2k - b & -k \\ -2b & 4 - 4b - 2k \end{pmatrix}$$

Consider the equilibrium point $(0, 0)$

$$J(0, 0) = \begin{pmatrix} 3 - 2(0) - (0) & -(0) \\ -2(0) & 4 - 4(0) - 2(0) \end{pmatrix}$$

$$J(0, 0) = \begin{pmatrix} 3 & 0 \\ 0 & 4 \end{pmatrix}$$

Looking for solutions of the form $\mathbf{x}(t) = \xi e^{rt}$ gives the linear system

$$\begin{pmatrix} 3 - r & 0 \\ 0 & 4 - r \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This system of equations has a nonzero solution if and only if the determinant of the coefficient matrix is zero, that is

$$\begin{vmatrix} 3 - r & 0 \\ 0 & 4 - r \end{vmatrix} = 0$$

$$(3 - r)(4 - r) = 0$$

$$3 - r = 0 \quad \text{or} \quad 4 - r = 0$$

$$r_1 = 3 \quad r_2 = 4$$

Therefore, the eigenvalues are $r_1 = 3$ and $r_2 = 4$. As these are two distinct real eigenvalues that are of the same sign, this point is a **node**. Moreover, as both eigenvalues are positive, the node is **unstable**.

For $r_1 = 3$, the eigenvector must satisfy

$$\begin{pmatrix} 3-3 & 0 \\ 0 & 4-3 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the second row

$$0\xi_1 + \xi_2 = 0$$

$$\xi_2 = 0$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} \xi_1 \\ 0 \end{pmatrix} = \xi_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_1 = 3$ is $\xi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

For $r_2 = 4$, the eigenvector must satisfy

$$\begin{pmatrix} 3-4 & 0 \\ 0 & 4-4 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the first row

$$-\xi_1 + 0\xi_2 = 0$$

$$-\xi_1 = 0$$

$$\xi_1 = 0$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \xi_2 \end{pmatrix} = \xi_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_2 = 4$ is $\xi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Near to (0,0) the solution has the form

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)}$$

$$\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{4t}$$

Consider the equilibrium point (0,2)

$$J(0,2) = \begin{pmatrix} 3 - 2(0) - (2) & -(0) \\ -2(2) & 4 - 4(2) - 2(0) \end{pmatrix}$$

$$J(0,2) = \begin{pmatrix} 1 & 0 \\ -4 & -4 \end{pmatrix}$$

Looking for solutions of the form $\mathbf{x}(t) = \xi e^{rt}$ gives the linear system

$$\begin{pmatrix} 1-r & 0 \\ -4 & -4-r \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This system of equations has a nonzero solution if and only if the determinant of the coefficient matrix is zero, that is

$$\begin{vmatrix} 1-r & 0 \\ -4 & -4-r \end{vmatrix} = 0$$

$$(1-r)(-4-r) - (0)(-4) = 0$$

$$(1-r)(-4-r) = 0$$

$$1-r=0 \quad \text{or} \quad -4-r=0$$

$$r_1 = 1 \quad r_2 = -4$$

Therefore, the eigenvalues are $r_1 = 1$ and $r_2 = -4$. As these are two distinct real eigenvalues that are of the opposite sign, this point is a **saddle**. Saddle points are always **unstable**.

For $r_1 = 1$, the eigenvector must satisfy

$$\begin{pmatrix} 1-1 & 0 \\ -4 & -4-1 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ -4 & -5 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the second row

$$-4\xi_1 - 5\xi_2 = 0$$

$$-4\xi_1 = 5\xi_2$$

$$\xi_1 = -\frac{5}{4}\xi_2$$

$$\text{let } \xi_2 = -4$$

$$\xi_1 = -\frac{5}{4}(-4)$$

$$\xi_1 = 5$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_1 = 1$ is $\xi^{(1)} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}$

For $r_2 = -4$, the eigenvector must satisfy

$$\begin{pmatrix} 1 - (-4) & 0 \\ -4 & -4 - (-4) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 0 \\ -4 & 0 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the first row

$$5\xi_1 + 0\xi_2 = 0$$

$$5\xi_1 = 0$$

$$\xi_1 = 0$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \xi_2 \end{pmatrix} = \xi_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_2 = -4$ is $\xi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Near to (0,2) the solution has the form

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)}$$

$$\mathbf{x} = c_1 \begin{pmatrix} 5 \\ -4 \end{pmatrix} e^t + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-4t}$$

Consider the equilibrium point (3,0)

$$J(3,0) = \begin{pmatrix} 3 - 2(3) - (0) & -(3) \\ -2(0) & 4 - 4(0) - 2(3) \end{pmatrix}$$

$$J(3,0) = \begin{pmatrix} -3 & -3 \\ 0 & -2 \end{pmatrix}$$

Looking for solutions of the form $\mathbf{x}(t) = \xi e^{rt}$ gives the linear system

$$\begin{pmatrix} -3 - r & -3 \\ 0 & -2 - r \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This system of equations has a nonzero solution if and only if the determinant of the coefficient matrix is zero, that is

$$\begin{vmatrix} -3 - r & -3 \\ 0 & -2 - r \end{vmatrix} = 0$$

$$(-3 - r)(-2 - r) - (-3)(0) = 0$$

$$(-3 - r)(-2 - r) = 0$$

$$-3 - r = 0 \quad \text{or} \quad -2 - r = 0$$

$$r_1 = -3$$

$$r_2 = -2$$

Therefore, the eigenvalues are $r_1 = -3$ and $r_2 = -2$. As these are two distinct real eigenvalues that are of the same sign, this point is a **node**. Moreover, as both eigenvalues are negative, the node is **stable**.

For $r_1 = -3$, the eigenvector must satisfy

$$\begin{pmatrix} -3 - (-3) & -3 \\ 0 & -2 - (-3) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the second row

$$0\xi_1 + \xi_2 = 0$$

$$\xi_2 = 0$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} \xi_1 \\ 0 \end{pmatrix} = \xi_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_1 = -3$ is $\xi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

For $r_2 = -2$, the eigenvector must satisfy

$$\begin{pmatrix} -3 - (-2) & -3 \\ 0 & -2 - (-2) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & -3 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the first row

$$-\xi_1 - 3\xi_2 = 0$$

$$\xi_1 = -3\xi_2$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} -3\xi_2 \\ \xi_2 \end{pmatrix} = \xi_2 \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_2 = -2$ is $\xi^{(2)} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$

Near to (3,0) the solution has the form

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)}$$

$$\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} -3 \\ 1 \end{pmatrix} e^{-2t}$$

- (c) To discuss the trajectories in the neighbourhood of each of the critical points, mini phase planes were sketched.

Firstly, near the equilibrium point (0,0)

$$\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{4t}$$

Finding k nullcline

$$\frac{dk}{dt} = 3k$$

$$\text{let } \frac{dk}{dt} = 0$$

$$3k = 0$$

$$k = 0$$

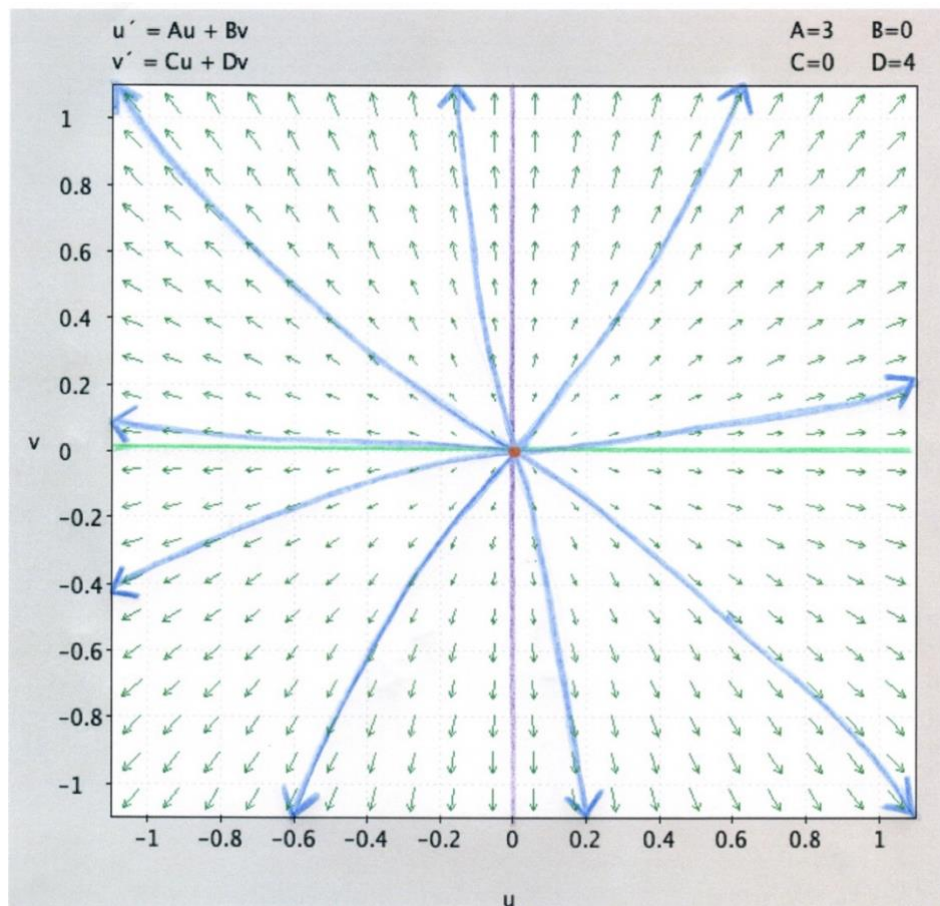
Finding b nullcline

$$\frac{db}{dt} = 4b$$

$$\text{let } \frac{db}{dt} = 0$$

$$4b = 0$$

$$b = 0$$



— $\mathbf{x}^{(1)}(t)$
— $\mathbf{x}^{(2)}(t)$
— $\frac{dk}{dt} = 0$
— $\frac{db}{dt} = 0$

The equilibrium point located at the origin (0,0) indicates the extinction of both species. In the neighbourhood of this equilibrium point, the trajectories flow from the node and are tangential to the eigenvector associated with the smaller eigenvalue. Therefore, this can be referred to as a source point (or a repelling node).

Secondly, near the equilibrium point (0,2)

$$\mathbf{x} = c_1 \begin{pmatrix} 5 \\ -4 \end{pmatrix} e^t + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-4t}$$

Finding k nullcline

$$\frac{dk}{dt} = k$$

$$\text{let } \frac{dk}{dt} = 0$$

$$k = 0$$

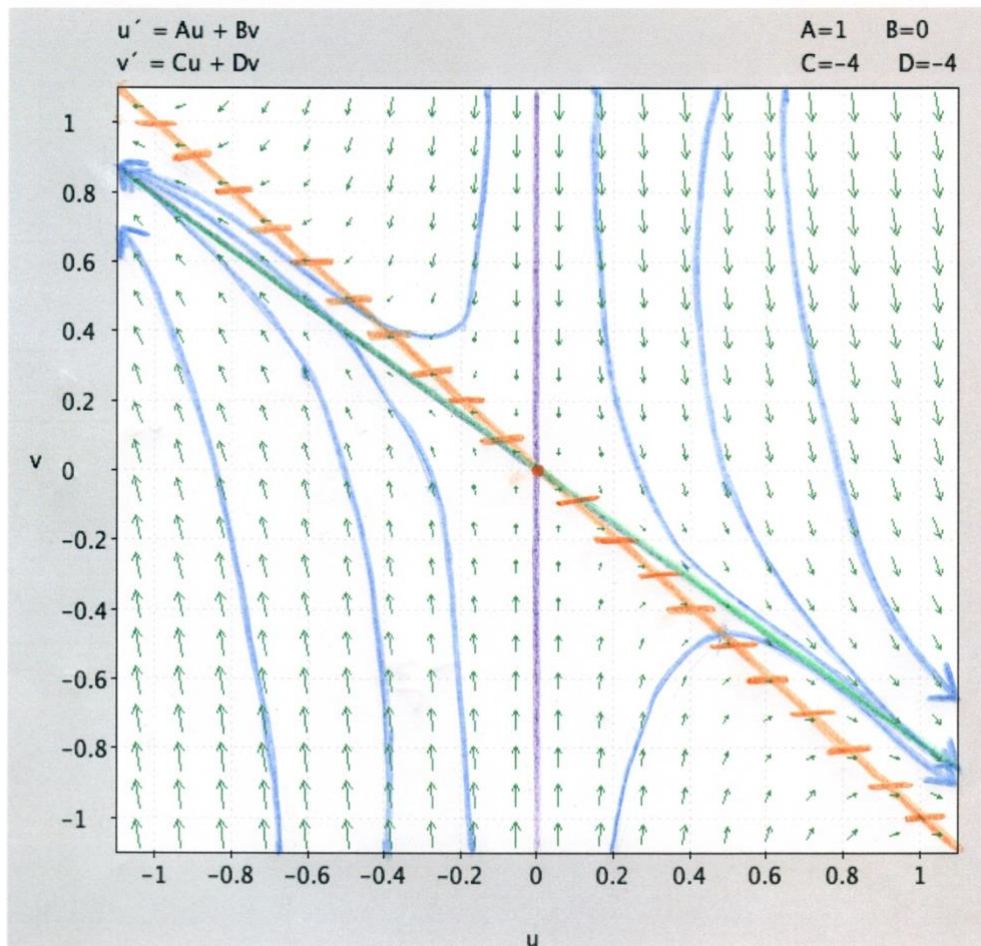
Finding b nullcline

$$\frac{db}{dt} = -4k - 4b$$

$$\text{let } \frac{db}{dt} = 0$$

$$-4k - 4b = 0$$

$$b = -k$$



— $\mathbf{x}^{(1)}(t)$
— $\mathbf{x}^{(2)}(t)$
— $\frac{dk}{dt} = 0$
— $\frac{db}{dt} = 0$

The equilibrium point located at (0,2) indicates the extinction of the drop bears. In the neighbourhood of this equilibrium point, almost all the trajectories depart from the saddle point as t increases, and approach asymptotically the eigenvector associated with the positive eigenvalue.

Thirdly, near the equilibrium point (3,0)

$$\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-3t} + c_2 \begin{pmatrix} -3 \\ 1 \end{pmatrix} e^{-2t}$$

Finding k nullcline

$$\frac{dk}{dt} = -3k - 3b$$

$$\text{let } \frac{dk}{dt} = 0$$

$$-3k - 3b = 0$$

$$k = -b$$

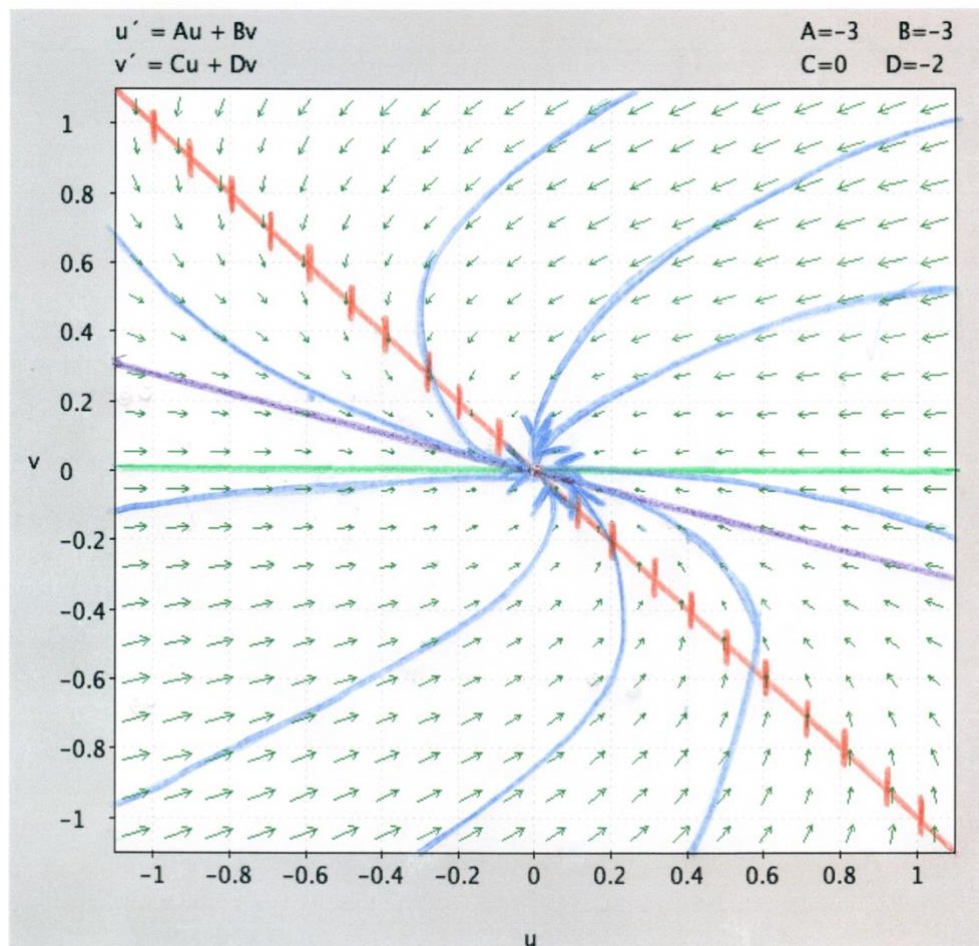
Finding b nullcline

$$\frac{db}{dt} = -2b$$

$$\text{let } \frac{db}{dt} = 0$$

$$-2b = 0$$

$$b = 0$$



— $\mathbf{x}^{(1)}(t)$
— $\mathbf{x}^{(2)}(t)$
— $\frac{dk}{dt} = 0$
— $\frac{db}{dt} = 0$

The equilibrium point located at (3,0) indicates the extinction of the bunyips. In the neighbourhood of this equilibrium point, the trajectories flow towards the node and are tangential to the eigenvector associated with the numerically smaller eigenvalue. Therefore, this can be referred to as a sink (or an attracting node).

- (d) A phase plane of the entire system was produced by generating the directional fields using a computational program, and subsequently annotating them by hand to show the key features of the system and example trajectories for each region.

Finding k nullcline

$$\frac{dk}{dt} = k(3 - k - b)$$

$$\text{let } \frac{dk}{dt} = 0$$

$$k(3 - k - b) = 0$$

$$k = 0 \text{ or } k = 3 - b$$

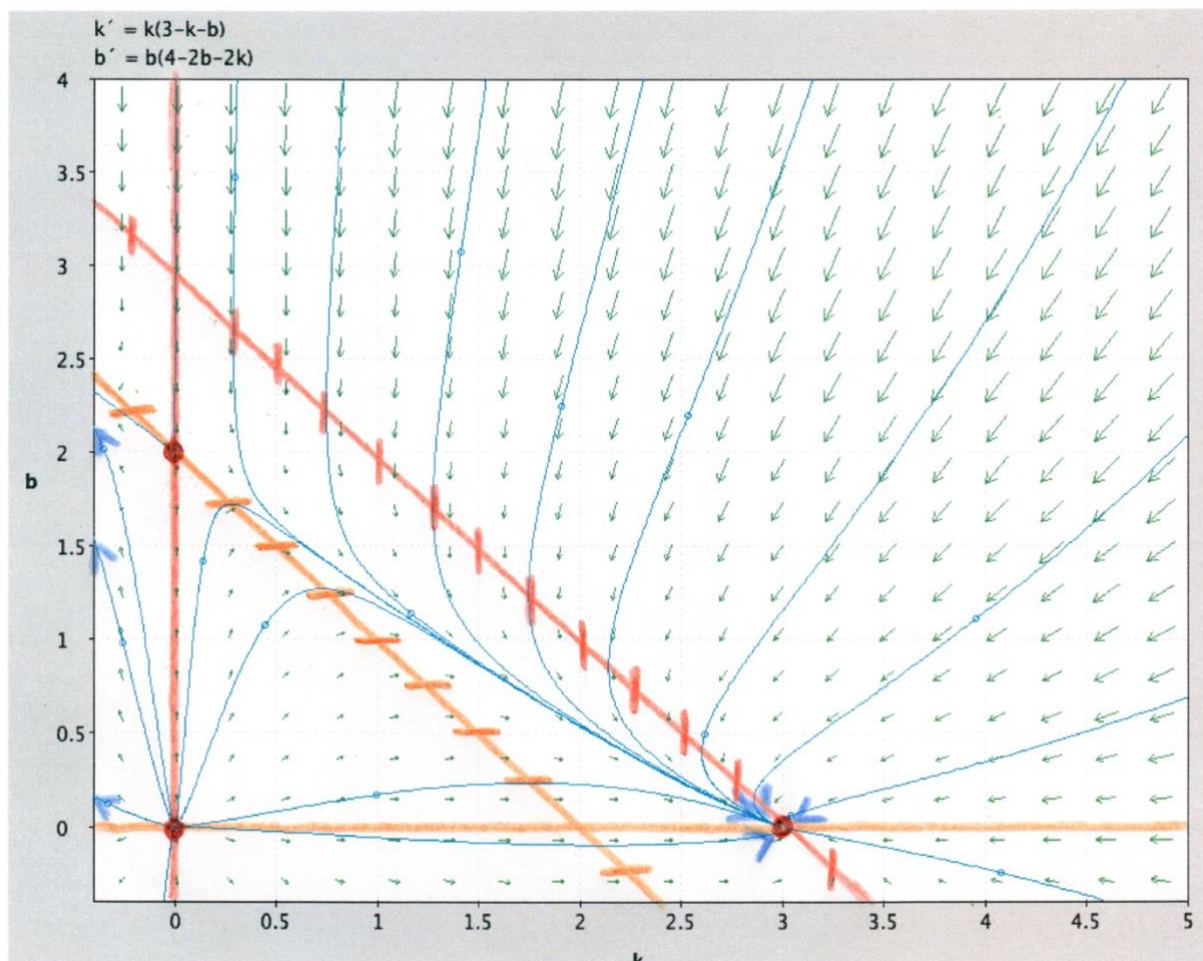
Finding b nullcline

$$\frac{db}{dt} = b(4 - 2b - 2k)$$

$$\text{let } \frac{db}{dt} = 0$$

$$b(4 - 2b - 2k) = 0$$

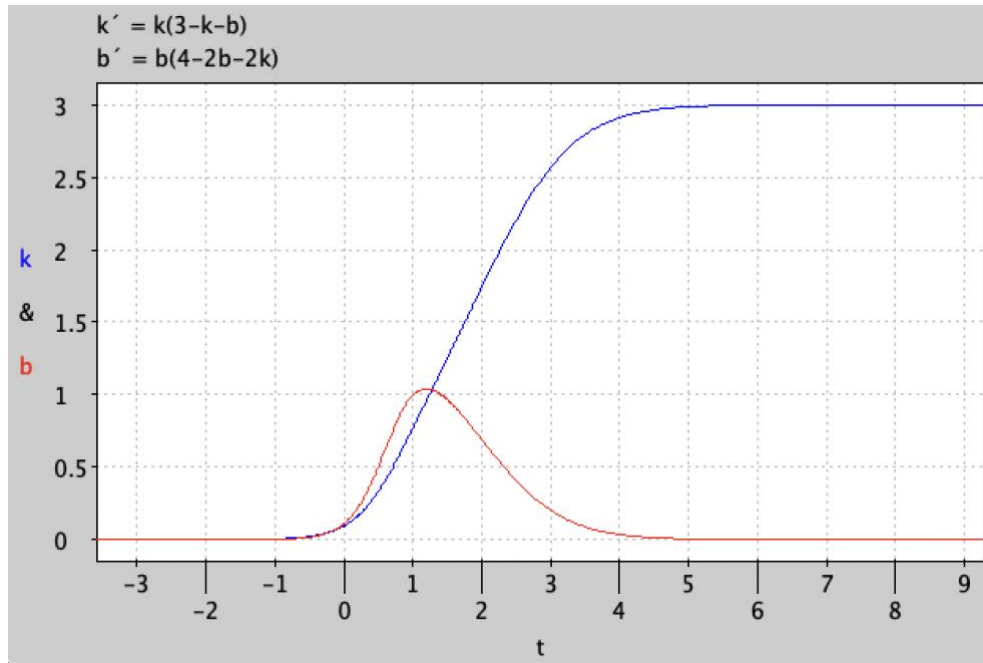
$$b = 0 \text{ or } b = 2 - k$$



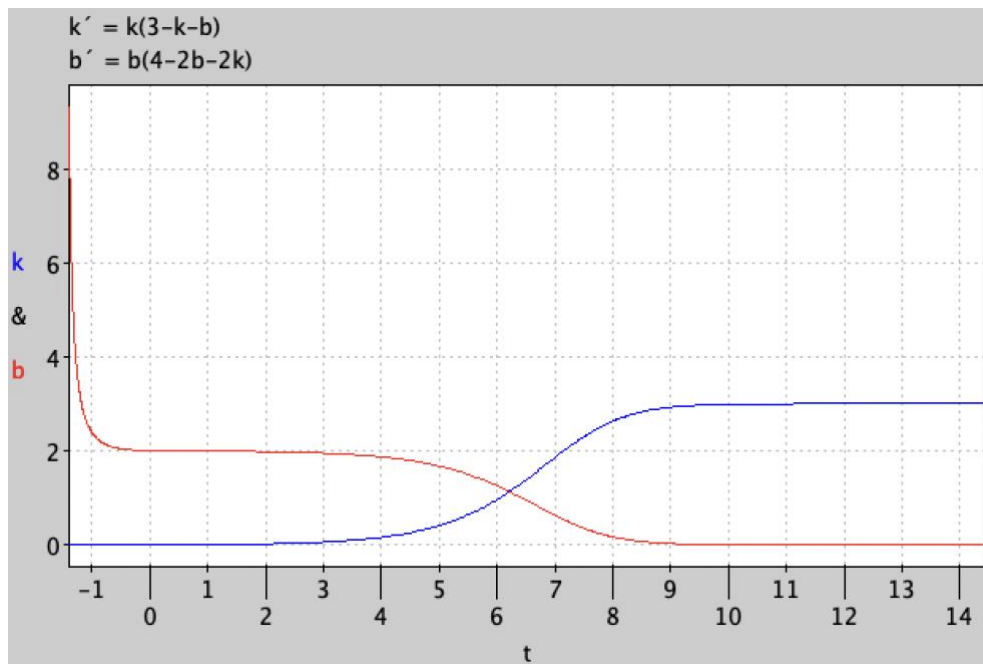
$$\text{---} \frac{dk}{dt} = 0 \quad \text{---} \frac{db}{dt} = 0$$

The phase plane is a useful geometric way of viewing the solutions (trajectories) from many initial conditions. Each trajectory denotes a solution associated with a particular initial condition, and each has an arrow to show the direction of evolution as t increases. The nullclines show the points where k is independent of time t and the points where b is also no longer changing with time. The intersection of any two nullclines represent steady-state values of fixed equilibrium points of the system.

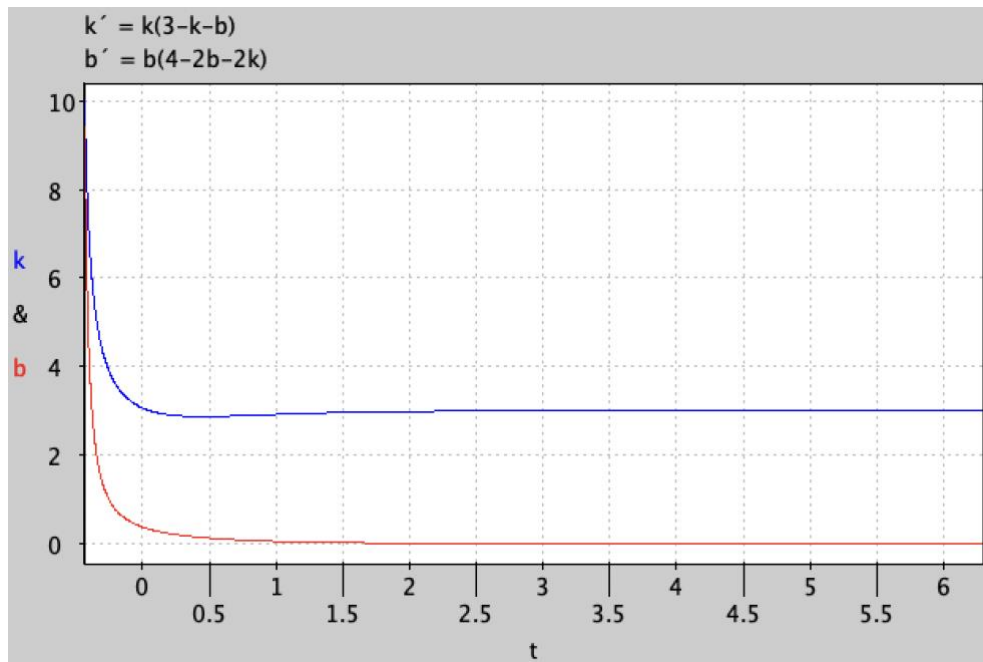
- (e) As the scaled competition model to describe the population dynamics of drop bears and bunyips are a pair of nonlinear coupled differential equations, then the local behaviour near a critical point is nearly the same as for the linear case. However, away from the critical points, the behaviour can become much more complex. The behaviour of the drop bear and bunyip populations as $t \rightarrow \infty$ can be described below.



The plot above displays the time evolution of the drop bear and bunyip populations near the unstable node $(0,0)$ for a single trajectory. As $t \rightarrow \infty$, $k \rightarrow 3$ and $b \rightarrow 0$.



The plot above displays the time evolution of the drop bear and bunyip populations near the unstable saddle point $(0,2)$ for a single trajectory. As $t \rightarrow \infty$, $k \rightarrow 3$ and $b \rightarrow 0$.



The plot above displays the time evolution of the drop bear and bunyip populations near the stable node $(3,0)$ for a single trajectory. As $t \rightarrow \infty$, $k \rightarrow 3$ and $b \rightarrow 0$.

Therefore near all the equilibrium points, the population of the drop bear species is approaching 3, while the population of the bunyip species is heading towards extinction.

- (f) Assuming the model correctly captures the population dynamics, the following advice can be given to Moira to uphold her goal of protecting both species. The bunyips are an endangered species that will become extinct soon as their population is declining. The competitive exclusion principle says that two species can't stably coexist in a habitat if they are competing for identical resources. Thus, the competitive exclusion principle can be avoided if one or both of the competing drop bear and bunyip evolves to use a different resource, occupy a different area of the region, or feed during a different time of day. The result of this kind of evolution is that the drop bears and bunyips will use largely non-overlapping resources. This idea is called resource partitioning, and it helps the drop bears and bunyips coexist as there is less direct competition between them.

Question 2

The Lotka-Volterra predator prey equations are derived under the following assumptions:

- In the absence of predators, the prey grows without bound.
- In the absence of prey, the predators die out.
- Predators consume prey at a rate proportional to the number of encounters (predators only eat prey, nothing else).
- A fixed proportion of prey is killed in each encounter due to predation.
- The rate of predator growth is enhanced by a factor proportional to the amount of prey consumed.

This results in the model,

$$\frac{df}{dt} = af - bfm \quad a, b > 0$$

$$\frac{dm}{dt} = -cm + dfm \quad c, d > 0$$

The constants a, b, c, d are positive real parameters describing the interaction of the two species. These constants are typically based on empirical observations and depend on the particular species being studied.

This model derived above can be extended to explore the effects of an insecticide on the population of microbats (predator) and mosquitoes (prey). The following assumptions can now be made regarding what happens when an insecticide is employed.

- The insecticide harms both the insect and microbat species.
- The insect and microbat species die at different rates due to the effects of the insecticide.

Thus the model becomes,

Let $f(t)$ be the population of mosquitoes at time t

Let $m(t)$ be the population of microbats at time t

$$\frac{df}{dt} = af - bfm - \gamma f = (a - \gamma)f - bfm \quad a, b, \gamma > 0$$

$$\frac{dm}{dt} = -cm + dfm - \kappa m = -(c + \kappa)m + dfm \quad c, d, \kappa > 0$$

Where:

a is the natural birth rate of the mosquitoes in the absence of predation

b is the efficiency of the microbats to capture mosquitoes per encounter

c is the natural death rate of the microbats in the absence of food (mosquitoes)

d is the efficiency of converting predated mosquitoes into microbats

γ is the rate at which the insecticide kills the insects

κ is the rate at which the insecticide kills the microbats

This model is a first-order, nonlinear, homogenous, autonomous, coupled system of ordinary differential equations.

- Nonlinear, because of the $f \times m$ terms
- Homogeneous, because there are no terms that don't involve either f or m
- Autonomous, because the independent variable t does not appear explicitly on the right-hand side of the equations

To find the critical points, set each equation of the model equal to zero, i.e.,

$$\text{Let } \frac{df}{dt} = 0$$

$$af - bfm - \gamma f = 0$$

$$f(a - bm - \gamma) = 0$$

$$\therefore f = 0 \quad \text{or} \quad a - bm - \gamma = 0$$

$$a - bm = \gamma$$

$$-bm = -a + \gamma$$

$$m = \frac{-(a - \gamma)}{-b}$$

$$m = \frac{a - \gamma}{b}$$

$$\text{Let } \frac{dm}{dt} = 0$$

$$-cm + dfm - \kappa m = 0$$

$$m(df - c - \kappa) = 0$$

Firstly, if $f = 0$

$$m(d(0) - c - \kappa) = 0$$

$$m(-c - \kappa) = 0$$

$$m = 0$$

$\therefore (f, m) = (0, 0)$ is a critical point

Next, if $m = \frac{a - \gamma}{b}$

$$\frac{a - \gamma}{b}(df - c - \kappa) = 0$$

$$df - c - \kappa = 0$$

$$df = c + \kappa$$

$$f = \frac{c + \kappa}{d}$$

$\therefore (f, m) = \left(\frac{c + \kappa}{d}, \frac{a - \gamma}{b}\right)$ is a critical point

Therefore, there are two critical points: $(0, 0)$ and $\left(\frac{c + \kappa}{d}, \frac{a - \gamma}{b}\right)$.

To investigate what solutions look like near these critical points, the Jacobian Matrix is needed. This is because near the critical point, the phase plane for the nonlinear system will be similar to that of the linear system $\mathbf{x}' = \mathbf{J}\mathbf{x}$.

Finding the Jacobian matrix of the system

$$J(f, m) = \begin{pmatrix} \frac{\partial F}{\partial f}(f, m) & \frac{\partial F}{\partial m}(f, m) \\ \frac{\partial G}{\partial f}(f, m) & \frac{\partial G}{\partial m}(f, m) \end{pmatrix} \quad \begin{aligned} F &= af - bfm - \gamma f \\ G &= -cm + dfm - \kappa m \end{aligned}$$

$$J(f, m) = \begin{pmatrix} \frac{\partial}{\partial f}(af - bfm - \gamma f) & \frac{\partial}{\partial m}(af - bfm - \gamma f) \\ \frac{\partial}{\partial f}(-cm + dfm - \beta m) & \frac{\partial}{\partial m}(-cm + dfm - \kappa m) \end{pmatrix}$$

$$J(f, m) = \begin{pmatrix} a - bm - \gamma & -bf \\ dm & -c + df - \kappa \end{pmatrix}$$

The classification and stability of the critical points of the system with constants $a, b, c, d, \gamma, \kappa$ can be determined by substituting in these values into the Jacobian and finding the eigenvalues.

The following parameter values were chosen:

Let $a = 1.25$,

$b = 0.5$

$c = 0.6$

$d = 0.2$

$\gamma = 0.75$

$\kappa = 0.4$

$$\begin{aligned} J(f, m) &= \begin{pmatrix} 1.25 - (0.5)m - 0.75 & -(0.5)f \\ (0.2)m & -(0.6) + (0.2)f - 0.4 \end{pmatrix} \\ &= \begin{pmatrix} 1.25 - 0.5m - 0.75 & -0.5f \\ 0.2m & -0.6 + 0.2f - 0.4 \end{pmatrix} \end{aligned}$$

$$\therefore J(f, m) = \begin{pmatrix} 0.5 - 0.5m & -0.5f \\ 0.2m & 0.2f - 1 \end{pmatrix}$$

For the critical point (0,0)

$$J(0,0) = \begin{pmatrix} 0.5 - 0.5(0) & -0.5(0) \\ 0.2(0) & 0.2(0) - 1 \end{pmatrix}$$

$$J(0,0) = \begin{pmatrix} 0.5 & 0 \\ 0 & -1 \end{pmatrix}$$

Looking for solutions of the form $\mathbf{x}(t) = \xi e^{rt}$ gives the linear system

$$\begin{pmatrix} 0.5 - r & 0 \\ 0 & -1 - r \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This system of equations has a nonzero solution if and only if the determinant of the coefficient matrix is zero, that is

$$\begin{vmatrix} 0.5 - r & 0 \\ 0 & -1 - r \end{vmatrix} = 0$$

$$(0.5 - r)(-1 - r) = 0$$

$$0.5 - r = 0 \quad \text{or} \quad -1 - r = 0$$

$$r_1 = \frac{1}{2} \quad r_2 = -1$$

Therefore, the eigenvalues are $r_1 = \frac{1}{2}$ and $r_2 = -1$. As these are two distinct real eigenvalues that are of the opposite sign, this point is a **saddle**. Saddle points are always **unstable**.

For $r_1 = \frac{1}{2}$, the eigenvector must satisfy

$$\begin{pmatrix} 0.5 - 0.5 & 0 \\ 0 & -1 - 0.5 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & -1.5 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the second row

$$0\xi_1 - \frac{3}{2}\xi_2 = 0$$

$$-\frac{3}{2}\xi_2 = 0$$

$$\xi_2 = 0$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} \xi_1 \\ 0 \end{pmatrix} = \xi_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_1 = \frac{1}{2}$ is $\xi^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

For $r_2 = -1$, the eigenvector must satisfy

$$\begin{pmatrix} 0.5 - (-1) & 0 \\ 0 & -1 - (-1) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1.5 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the first row

$$\frac{3}{2}\xi_1 + \xi_2 = 0$$

$$\frac{3}{2}\xi_1 = 0$$

$$\xi_1 = 0$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \xi_2 \end{pmatrix} = \xi_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_2 = -1$ is $\xi^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Near to (0,0) the solution has the form

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)} \quad f(t) = c_1 e^{\frac{1}{2}t}$$

$$\mathbf{x} = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{\frac{1}{2}t} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-t} \quad m(t) = c_2 e^{-t}$$

For the critical point $\left(\frac{c+\kappa}{d}, \frac{a-\gamma}{b}\right)$

$$\therefore \left(\frac{c+\kappa}{d}, \frac{a-\gamma}{b}\right) = \left(\frac{0.6+0.4}{0.2}, \frac{1.25-0.75}{0.5}\right) = (5, 1)$$

$$J(5,1) = \begin{pmatrix} 0.5 - 0.5(1) & -0.5(5) \\ 0.2(1) & 0.2(5) - 1 \end{pmatrix}$$

$$J(5,1) = \begin{pmatrix} 0 & -2.5 \\ 0.2 & 0 \end{pmatrix}$$

Looking for solutions of the form $\mathbf{x}(t) = \xi e^{rt}$ gives the linear system

$$\begin{pmatrix} 0-r & -2.5 \\ 0.2 & 0-r \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This system of equations has a nonzero solution if and only if the determinant of the coefficient matrix is zero, that is

$$\begin{vmatrix} -r & -2.5 \\ 0.2 & -r \end{vmatrix} = 0$$

$$(-r)^2 - (-2.5)(0.2) = 0$$

$$r^2 + \frac{1}{2} = 0$$

$$r^2 = -\frac{1}{2}$$

$$r = 0 \pm \sqrt{-\frac{1}{2}}$$

$$r = \pm \sqrt{\frac{1}{2}} \sqrt{-1}$$

$$r = \pm \sqrt{\frac{1}{2}} i$$

Therefore, the eigenvalues are $r_1 = \sqrt{\frac{1}{2}} i$ and $r_2 = -\sqrt{\frac{1}{2}} i$. As these are complex conjugate eigenvalues that are purely imaginary (real part equal to 0), this point is a **centre**. **Centres are stable**, but **not asymptotically stable**.

For $r_1 = \sqrt{\frac{1}{2}} i$, the eigenvector must satisfy

$$\begin{pmatrix} -\sqrt{\frac{1}{2}} i & -2.5 \\ 0.2 & -\sqrt{\frac{1}{2}} i \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the second row

$$\frac{1}{5} \xi_1 - \sqrt{\frac{1}{2}} i \xi_2 = 0$$

$$\frac{1}{5} \xi_1 = \sqrt{\frac{1}{2}} i \xi_2$$

$$\xi_1 = 5 \sqrt{\frac{1}{2}} i \xi_2$$

$$\text{let } \xi_2 = \frac{i}{5}$$

$$\xi_1 = 5 \sqrt{\frac{1}{2}} i \left(\frac{i}{5} \right)$$

$$\xi_1 = \sqrt{\frac{1}{2}} i^2$$

$$\xi_1 = -\sqrt{\frac{1}{2}}$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} -\sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_1 = \sqrt{\frac{1}{2}} i$ is $\xi^{(1)} = \begin{pmatrix} -\sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix}$

For $r_2 = -\sqrt{\frac{1}{2}}i$, the eigenvector must satisfy

$$\begin{pmatrix} -\left(-\sqrt{\frac{1}{2}}i\right) & -2.5 \\ 0.2 & -\left(-\sqrt{\frac{1}{2}}i\right) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} \sqrt{\frac{1}{2}}i & -2.5 \\ 0.2 & \sqrt{\frac{1}{2}}i \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Using the second row

$$\frac{1}{5}\xi_1 + \sqrt{\frac{1}{2}}i \xi_2 = 0$$

$$\frac{1}{5}\xi_1 = -\sqrt{\frac{1}{2}}i \xi_2$$

$$\xi_1 = -5\sqrt{\frac{1}{2}}i \xi_2$$

$$\text{let } \xi_2 = \frac{i}{5}$$

$$\xi_1 = -5\sqrt{\frac{1}{2}}i \left(\frac{i}{5}\right)$$

$$\xi_1 = -\sqrt{\frac{1}{2}}i^2$$

$$\xi_1 = \sqrt{\frac{1}{2}}$$

$$\therefore \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix}$$

Therefore the corresponding eigenvector to $r_2 = -\sqrt{\frac{1}{2}}i$ is $\xi^{(2)} = \begin{pmatrix} \sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix}$

Near to (5,1) the solution has the form

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)}$$

$$\mathbf{x} = c_1 \begin{pmatrix} -\sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix} e^{\sqrt{\frac{1}{2}}it} + c_2 \begin{pmatrix} \sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix} e^{-\sqrt{\frac{1}{2}}it}$$

This general solution above is complex, However, real valued solutions are needed.

Using $\mathbf{x}^{(1)}(t)$

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} -\sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix} e^{i\left(\sqrt{\frac{1}{2}}t\right)}$$

Using Euler's Formula: $e^{ix} = \cos x + i \sin x$

$$\begin{aligned} \therefore \mathbf{x}^{(1)}(t) &= \begin{pmatrix} -\sqrt{\frac{1}{2}} \\ \frac{i}{5} \end{pmatrix} \left[\cos\left(\sqrt{\frac{1}{2}}t\right) + i \sin\left(\sqrt{\frac{1}{2}}t\right) \right] \\ &= \begin{pmatrix} -\sqrt{\frac{1}{2}} \cos\left(\sqrt{\frac{1}{2}}t\right) - \sqrt{\frac{1}{2}} i \sin\left(\sqrt{\frac{1}{2}}t\right) \\ \frac{i}{5} \cos\left(\sqrt{\frac{1}{2}}t\right) + \frac{i^2}{5} \sin\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix} \\ &= \begin{pmatrix} -\sqrt{\frac{1}{2}} \cos\left(\sqrt{\frac{1}{2}}t\right) - \sqrt{\frac{1}{2}} i \sin\left(\sqrt{\frac{1}{2}}t\right) \\ \frac{i}{5} \cos\left(\sqrt{\frac{1}{2}}t\right) - \frac{1}{5} \sin\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix} \\ &= \begin{pmatrix} -\sqrt{\frac{1}{2}} \cos\left(\sqrt{\frac{1}{2}}t\right) \\ -\frac{1}{5} \sin\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix} + i \begin{pmatrix} -\sqrt{\frac{1}{2}} \sin\left(\sqrt{\frac{1}{2}}t\right) \\ \frac{1}{5} \cos\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix} \end{aligned}$$

The real valued solutions are therefore

$$\mathbf{u} = \begin{pmatrix} -\sqrt{\frac{1}{2}} \cos\left(\sqrt{\frac{1}{2}}t\right) \\ -\frac{1}{5} \sin\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} -\sqrt{\frac{1}{2}} \sin\left(\sqrt{\frac{1}{2}}t\right) \\ \frac{1}{5} \cos\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix}$$

$$\mathbf{x}(t) = c_1 \mathbf{u}(t) + c_2 \mathbf{v}(t)$$

$$\mathbf{x}(t) = c_1 \begin{pmatrix} -\sqrt{\frac{1}{2}} \cos\left(\sqrt{\frac{1}{2}}t\right) \\ -\frac{1}{5} \sin\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix} + c_2 \begin{pmatrix} -\sqrt{\frac{1}{2}} \sin\left(\sqrt{\frac{1}{2}}t\right) \\ \frac{1}{5} \cos\left(\sqrt{\frac{1}{2}}t\right) \end{pmatrix}$$

The domain was implicitly shifted so that the critical point was at the origin. Therefore to undo this, the critical point needs to be added back on.

$$f(t) = -c_1 \sqrt{\frac{1}{2}} \cos\left(\sqrt{\frac{1}{2}}t\right) - c_2 \sqrt{\frac{1}{2}} \sin\left(\sqrt{\frac{1}{2}}t\right) + 5$$

$$m(t) = -\frac{c_1}{5} \sin\left(\sqrt{\frac{1}{2}}t\right) + \frac{c_2}{5} \cos\left(\sqrt{\frac{1}{2}}t\right) + 1$$

To determine how conclusions are dependent on the choice of parameter values, the model was explored in general near the critical points, rather than substituting specific values in for the constants $a, b, c, d, \gamma, \kappa$.

$$J(f, m) = \begin{pmatrix} a - bm - \gamma & -bf \\ dm & -c + df - \kappa \end{pmatrix}$$

For the critical point (0,0)

$$J(0,0) = \begin{pmatrix} a - b(0) - \gamma & -b(0) \\ d(0) & -c + d(0) - \kappa \end{pmatrix}$$

$$J(0,0) = \begin{pmatrix} a - \gamma & 0 \\ 0 & -c - \kappa \end{pmatrix}$$

$$\begin{vmatrix} a - \gamma - r & 0 \\ 0 & -c - \kappa - r \end{vmatrix} = 0$$

$$(a - \gamma - r)(-c - \kappa - r) = 0$$

$$a - \gamma - r = 0 \quad \text{or} \quad -c - \kappa - r = 0$$

$$r_1 = a - \gamma \quad r_2 = -c - \kappa$$

For the critical point $\left(\frac{c + \kappa}{d}, \frac{a - \gamma}{b}\right)$

$$J\left(\frac{c + \kappa}{d}, \frac{a - \gamma}{b}\right) = \begin{pmatrix} a - b\left(\frac{a - \gamma}{b}\right) - \gamma & -b\left(\frac{c + \kappa}{d}\right) \\ d\left(\frac{a - \gamma}{b}\right) & -c + d\left(\frac{c + \kappa}{d}\right) - \kappa \end{pmatrix}$$

$$J\left(\frac{c + \kappa}{d}, \frac{a - \gamma}{b}\right) = \begin{pmatrix} 0 & \frac{-b(c + \kappa)}{d} \\ \frac{d(a - \gamma)}{b} & 0 \end{pmatrix}$$

$$\begin{vmatrix} -r & \frac{-b(c + \kappa)}{d} \\ \frac{d(a - \gamma)}{b} & -r \end{vmatrix} = 0$$

$$(-r)^2 - \left(\frac{-b(c + \kappa)}{d}\right)\left(\frac{d(a - \gamma)}{b}\right) = 0$$

$$r^2 - [-(c + \kappa)(a - \gamma)] = 0$$

$$r^2 + (c + \kappa)(a - \gamma) = 0$$

$$r^2 = -(c + \kappa)(a - \gamma)$$

$$r = \pm\sqrt{-(c + \kappa)(a - \gamma)}$$

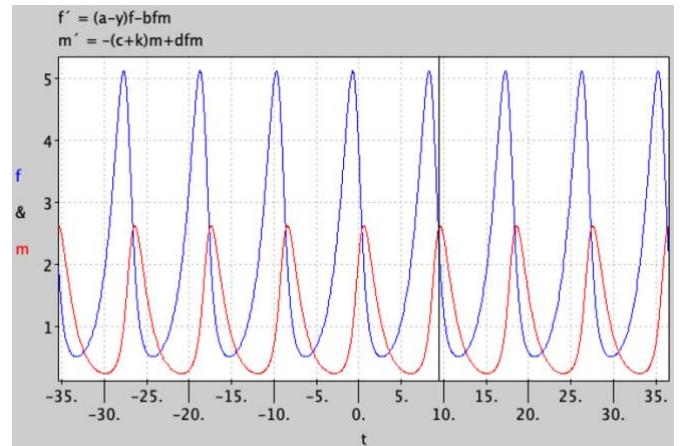
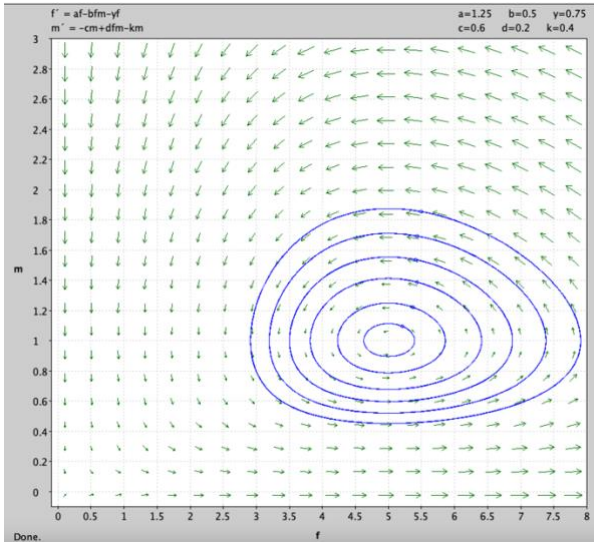
$$r = \pm\sqrt{(-a + \gamma)(c + \kappa)}$$

The magnitude of the parameters, relative to each other, determine the classification and hence, stability of the critical points. The types of critical point can be identified from the eigenvalues solved above for general parameter values. The results of this analysis can be summarised in the table below:

Critical Point	<i>Near</i> (0,0)	<i>Near</i> $\left(\frac{c+\kappa}{d}, \frac{a-\gamma}{b}\right)$
Jacobian	$\begin{pmatrix} a-\gamma & 0 \\ 0 & -c-\kappa \end{pmatrix}$	$\begin{pmatrix} 0 & \frac{-bc-b\kappa}{d} \\ \frac{ad-\gamma d}{b} & 0 \end{pmatrix}$
Eigenvalues	$a-\gamma, -c-\kappa$	$\sqrt{(-a+\gamma)(c+\kappa)}, -\sqrt{(-a+\gamma)(c+\kappa)}$
Type	<i>Saddle when $a > \gamma$ & $c \geq \kappa$</i> <i>Saddle when $a > \gamma$ & $c \leq \kappa$</i> <i>Sink when $a < \gamma$ & $c \geq \kappa$</i> <i>Sink when $a < \gamma$ & $c \leq \kappa$</i>	<i>Centre when $a > \gamma$ & $c \geq \kappa$</i> <i>Centre when $a > \gamma$ & $c \leq \kappa$</i> <i>Saddle when $a < \gamma$ & $c \geq \kappa$</i> <i>Saddle when $a < \gamma$ & $c \leq \kappa$</i>

Moreover, the following conclusions about the populations can be made:

- The saddle point is unstable, which means as long as there is no point where the population drops to zero, it will rebound even if it gets small.
- The critical point located at $\left(\frac{c+\beta}{d}, \frac{a-\gamma}{b}\right)$ is a centre (as the real part of the eigenvalues is zero), thus it is stable and the trajectories form a closed path and travel in closed elliptical orbits around the critical point (see plot below). However, it is not an asymptotically stable critical point, as the trajectories never approach the critical point, at least if one starts sufficiently close to the critical point, staying close to the critical point is ensured.
- The microbat population achieves its maximum value after the mosquito population achieves its maximum; the prey population leads the predator population (see plot below).
- The amplitudes of the oscillations depend on the initial conditions. Thus, starting with larger initial populations for both the predator and prey results in being on a larger loop than starting closer to the critical value.



As this extended model to explore the effects of an insecticide has the same form as the original Lotka-Volterra system when a is replaced by $a - \gamma > 0$ and c is replaced by $c + \kappa$, this new situation has a period of

$$T = \frac{2\pi}{\sqrt{(a - \gamma)(c + \kappa)}}$$

Therefore, the mosquito and microbat population vary cyclically with this period, which is independent of the initial conditions.

Additionally, the average populations, which are independent of the initial conditions over a complete cycle are,

$$\bar{f} = \frac{c + \kappa}{d} \quad \text{and} \quad \bar{m} = \frac{a - \gamma}{b}$$

for the mosquitoes and microbats respectively. Hence, the average mosquito population increases, and the average predator population decreases. Therefore in the long run, using an insecticide that also eliminates some of the microbat predator species is counterproductive.

Question 3

General Form: $\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$

$$(a) \quad \cos(x) y''(x) + \cos(x) y'(x) + y(x) = 0$$

$$\frac{\cos(x) y''(x)}{\cos(x)} + \frac{\cos(x) y'(x)}{\cos(x)} + \frac{y(x)}{\cos(x)} = \frac{0}{\cos(x)}$$

$$y''(x) + y'(x) + \frac{1}{\cos(x)} y(x) = 0$$

$$\therefore P(x) = 1 \quad \text{and} \quad Q(x) = \frac{1}{\cos(x)}$$

Since $\lim_{x \rightarrow 0} P(x) = \lim_{x \rightarrow 0} 1 = 1$, and

$$\lim_{x \rightarrow 0} Q(x) = \lim_{x \rightarrow 0} \frac{1}{\cos(x)} = \frac{1}{\cos(0)} = \frac{1}{1} = 1$$

Both are finite

$\therefore x = 0$ is an ordinary point in this differential equation

$$(b) \quad x^2 y'' + \left(\frac{3}{x}\right) y' + \frac{y}{x^2} = 0$$

$$\frac{x^2 y''}{x^2} + \frac{\left(\frac{3}{x}\right) y'}{x^2} + \frac{y}{x^2} = \frac{0}{x^2}$$

$$y'' + \frac{3y'}{x \cdot x^2} + \frac{y}{x^2 \cdot x^2} = 0$$

$$y'' + \frac{3}{x^3} y' + \frac{1}{x^4} y = 0$$

$$\therefore P(x) = \frac{3}{x^3} \quad \text{and} \quad Q(x) = \frac{1}{x^4}$$

Since $\lim_{x \rightarrow 0} P(x) = \lim_{x \rightarrow 0} \frac{3}{x^3} = \frac{3}{(0)^3} = \infty$, and

$$\lim_{x \rightarrow 0} Q(x) = \lim_{x \rightarrow 0} \frac{1}{x^4} = \frac{1}{(0)^4} = \infty$$

$\therefore x = 0$ is a singular point

Further since $\lim_{x \rightarrow 0} x P(x) = \lim_{x \rightarrow 0} x \cdot \frac{3}{x^3} = \lim_{x \rightarrow 0} \frac{3}{x^2} = \frac{3}{(0)^2} = \infty$

$\therefore x = 0$ is an irregular singular point

$$(c) \quad (2x - 1)y'' + (2x^2 + x - 1)y' - (1 - 2x)y = 0$$

$$\frac{(2x - 1)y''}{(2x - 1)} + \frac{(2x^2 + x - 1)y'}{(2x - 1)} - \frac{(1 - 2x)y}{(2x - 1)} = \frac{0}{(2x - 1)}$$

$$y'' + \frac{(2x - 1)(x + 1)}{(2x - 1)}y' - \frac{-(2x - 1)}{(2x - 1)}y = 0$$

$$y'' + (x + 1)y' + y = 0$$

$$\therefore P(x) = (x + 1) \text{ and } Q(x) = 1$$

$$\text{Since } \lim_{x \rightarrow 0} P(x) = \lim_{x \rightarrow 0} (x + 1) = 0 + 1 = 1, \text{ and}$$

$$\lim_{x \rightarrow 0} Q(x) = \lim_{x \rightarrow 0} 1 = 1$$

Both are finite

$\therefore x = 0$ is an ordinary point in this differential equation

$$(d) \quad \frac{1}{t}s'' - \frac{1}{t+3}s' + \sin(t)s = 0$$

$$s'' - \frac{t}{t+3}s' + t \sin(t)s = 0$$

$$\therefore P(t) = -\frac{1}{t+3} \text{ and } Q(t) = t \sin(t)$$

$$\text{Since } \lim_{t \rightarrow 0} P(t) = \lim_{t \rightarrow 0} -\frac{1}{t+3} = \lim_{t \rightarrow 0} -\frac{0}{0+3} = -\frac{0}{3} = 0, \text{ and}$$

$$\lim_{t \rightarrow 0} Q(t) = \lim_{t \rightarrow 0} t \sin(t) = 0 \sin(0) = 0$$

Both are finite

$\therefore t = 0$ is an ordinary point in this differential equation

Question 4

Finding, and classifying, all the singular points in the differential equation

$$(x^2 - 1)y'' + xy' + 3y = 0$$

$$\frac{(x^2 - 1)y''}{(x^2 - 1)} + \frac{xy'}{(x^2 - 1)} + \frac{3y}{(x^2 - 1)} = \frac{0}{(x^2 - 1)}$$

$$y'' + \frac{x}{(x^2 - 1)}y' + \frac{3}{(x^2 - 1)}y = 0$$

$$\therefore P(x) = \frac{x}{(x^2 - 1)} \text{ and } Q(x) = \frac{3}{(x^2 - 1)}$$

Finding all the singular points

For singular points, $x^2 - 1 = 0$

$$x^2 = 1$$

$$x = \pm 1$$

$\therefore$ The singular points in the differential equation are $x = -1$ and $x = 1$

Verifying the found singular points

$$\text{Since } \lim_{x \rightarrow -1} P(x) = \lim_{x \rightarrow -1} \frac{x}{(x^2 - 1)} = \frac{-1}{(-1)^2 - 1} = \frac{-1}{0} = \infty, \text{ and}$$

$$\lim_{x \rightarrow -1} Q(x) = \lim_{x \rightarrow -1} \frac{3}{(x^2 - 1)} = \frac{3}{(-1)^2 - 1} = \frac{3}{0} = \infty$$

$$\text{Since } \lim_{x \rightarrow 1} P(x) = \lim_{x \rightarrow 1} \frac{1}{(x^2 - 1)} = \frac{1}{(1)^2 - 1} = \frac{1}{0} = \infty, \text{ and}$$

$$\lim_{x \rightarrow 1} Q(x) = \lim_{x \rightarrow 1} \frac{3}{(x^2 - 1)} = \frac{3}{(1)^2 - 1} = \frac{3}{0} = \infty$$

$\therefore x = -1$ and $x = 1$ are indeed singular points

*Classifying the singular points**For $x = -1$*

$$\begin{aligned}
 \text{Since } \lim_{x \rightarrow x_0} (x - x_0) P(x) &= \lim_{x \rightarrow -1} (x - (-1)) \frac{x}{(x^2 - 1)} \\
 &= \lim_{x \rightarrow -1} (x + 1) \frac{x}{(x - 1)(x + 1)} \\
 &= \lim_{x \rightarrow -1} \frac{x}{x - 1} \\
 &= \frac{-1}{-1 - 1} \\
 &= \frac{1}{2}, \text{ and}
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow x_0} (x - x_0)^2 Q(x) &= \lim_{x \rightarrow -1} (x - (-1))^2 \frac{3}{(x^2 - 1)} \\
 &= \lim_{x \rightarrow -1} (x + 1)^2 \frac{3}{(x - 1)(x + 1)} \\
 &= \lim_{x \rightarrow -1} \frac{3(x + 1)}{(x - 1)} \\
 &= \lim_{x \rightarrow -1} \frac{3x + 3}{x - 1} \\
 &= \frac{3(-1) + 3}{-1 - 1} \\
 &= \frac{0}{-2} \\
 &= 0
 \end{aligned}$$

Both are finite *$\therefore x = -1$ is a regular singular point in the differential equation**For $x = 1$*

$$\begin{aligned}
 \text{Since } \lim_{x \rightarrow x_0} (x - x_0) P(x) &= \lim_{x \rightarrow 1} (x - 1) \frac{x}{(x^2 - 1)} \\
 &= \lim_{x \rightarrow 1} (x - 1) \frac{x}{(x - 1)(x + 1)} \\
 &= \lim_{x \rightarrow 1} \frac{x}{x + 1} \\
 &= \frac{1}{1 + 1} \\
 &= \frac{1}{2}, \text{ and}
 \end{aligned}$$

$$\begin{aligned}\lim_{x \rightarrow x_0} (x - x_0)^2 Q(x) &= \lim_{x \rightarrow 1} (x - 1)^2 \frac{3}{(x^2 - 1)} \\&= \lim_{x \rightarrow 1} (x - 1)^2 \frac{3}{(x - 1)(x + 1)} \\&= \lim_{x \rightarrow 1} \frac{3(x - 1)}{(x + 1)} \\&= \lim_{x \rightarrow 1} \frac{3x - 3}{x + 1} \\&= \frac{3(1) - 3}{1 + 1} \\&= \frac{0}{2} \\&= 0\end{aligned}$$

Both are finite

$\therefore x = 1$ is also a regular singular point in the differential equation

Question 5

Considering the differential equation:

$$x(y''(x) - y(x)) + 2(y''(x) + y'(x) - y(x)) = 0$$

(a) To show that a regular singular point exists at $x = -2$

$$xy''(x) - xy(x) + 2y''(x) + 2y'(x) - 2y(x) = 0$$

$$(x + 2)y''(x) + 2y'(x) - (x + 2)y(x) = 0$$

$$\frac{(x + 2)y''(x)}{(x + 2)} + \frac{2y'(x)}{(x + 2)} - \frac{(x + 2)y(x)}{(x + 2)} = 0$$

$$y''(x) + \frac{2}{(x + 2)}y'(x) - y(x) = 0$$

$$\therefore P(x) = \frac{2}{(x + 2)} \quad \text{and} \quad Q(x) = -1$$

$$\text{Since } \lim_{x \rightarrow -2} P(x) = \lim_{x \rightarrow -2} \frac{2}{(x + 2)} = \frac{2}{-2 + 2} = \frac{2}{0} = \infty$$

$\therefore x = -2$ is a singular point

$$\begin{aligned} \text{Further since } \lim_{x \rightarrow x_0} (x - x_0) P(x) &= \lim_{x \rightarrow -2} (x - (-2)) \frac{2}{(x + 2)} \\ &= \lim_{x \rightarrow -2} (x + 2) \frac{2}{(x + 2)} \\ &= \lim_{x \rightarrow -2} 2 \\ &= 2, \quad \text{and} \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow x_0} (x - x_0)^2 Q(x) &= \lim_{x \rightarrow -2} (x - (-2))^2 (-1) \\ &= \lim_{x \rightarrow -2} (x + 2)^2 (-1) \\ &= \lim_{x \rightarrow -2} x^2 + 4x + 4 \\ &= (-2)^2 + 4(-2) + 4 \\ &= 4 - 8 + 4 \\ &= 0 \end{aligned}$$

Both are finite

$\therefore$ A regular singular point exists at $x = -2$

$$(b) (x+2)\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - (x+2)y = 0$$

New expressions are needed for $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of the new independent variable, z

$$\begin{aligned} \text{Let } z &= x - x_0 \\ &= x - (-2) \\ \mathbf{z} &= \mathbf{x + 2} \end{aligned}$$

Differentiating both sides w.r.t x

$$\begin{aligned} \frac{d}{dx}(z) &= \frac{d}{dx}(x+2) \\ \frac{dz}{dx} &= 1 \end{aligned}$$

Using the chain rule to relate x derivatives and z derivatives

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dz} \cdot \frac{dz}{dx} \\ &= \frac{dy}{dz}(1) \end{aligned}$$

$$\frac{dy}{dx} = \frac{dy}{dz}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx}\left(\frac{dy}{dx}\right) \\ &= \frac{d}{dz} \cdot \frac{dz}{dx} \left(\frac{dy}{dx}\right) \\ &= \frac{d}{dz} \cdot 1 \cdot \left(\frac{dy}{dz}\right) \end{aligned}$$

$$\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2}$$

Substitute into the d.e

$$(x+2)\frac{d^2y}{dz^2} + 2\frac{dy}{dz} - (x+2)y = 0$$

Replacing $x+2$ with z

$$z\frac{d^2y}{dz^2} + 2\frac{dy}{dz} - zy = 0$$

$$zy''(z) + 2y'(z) - zy(z) = 0$$

$$(c) \quad zy''(z) + 2y'(z) - zy(z) = 0$$

$$\frac{zy''(z)}{z} + \frac{2y'(z)}{z} - \frac{zy(z)}{z} = \frac{0}{z}$$

$$y''(z) + \frac{2}{z}y'(z) - y(z) = 0$$

$$\therefore P(z) = \frac{2}{z} \quad \text{and} \quad Q(z) = -1$$

Finding singular point(s)

$P(z)$ is analytic at all z except $z = 0$

$\therefore$ There is a singularity at $z = 0$

Verifying the found singular point

$$\text{Since } \lim_{z \rightarrow 0} P(z) = \lim_{z \rightarrow 0} \frac{2}{z} = \frac{2}{0} = \infty$$

$\therefore$ There is indeed a singularity at $z = 0$

Classifying the singular point

$$\text{Since } \lim_{z \rightarrow 0} z P(z) = \lim_{z \rightarrow 0} z \cdot \frac{2}{z} = \lim_{z \rightarrow 0} 2 = 2, \quad \text{and}$$

$$\lim_{z \rightarrow 0} z^2 Q(z) = \lim_{z \rightarrow 0} z^2 \cdot -1 = \lim_{z \rightarrow 0} -z^2 = -(0)^2 = 0$$

Both are finite

$\therefore z = 0$ is a regular singular point in the transformed differential equation

This corresponds to $x = -2$, (as $x = z - 2 = 0 - 2 = -2$),

which was the regular singular point shown earlier

All the values of $z \neq 0$ are the ordinary points of the d.e

Therefore, the Frobenius method is required to solve this differential equation as it is a series solution method used for solving differential equations when $x = 0$ (or in this case $z = 0$) is a regular singular point.

(d) Finding the appropriate p values in the Frobenius Form

As there is a regular singular point at $z = 0$, the Frobenius expansion can be used:

$$y(z) = z^p(a_0 + a_1z + a_2z^2 + a_3z^3 + \dots) = a_0z^p + a_1z^{p+1} + a_2z^{p+2} + a_3z^{p+3} + \dots$$

Step 1: Differentiate the Frobenius form

$$y(z) = a_0z^p + a_1z^{p+1} + a_2z^{p+2} + a_3z^{p+3} + \dots$$

$$\therefore y'(z) = pa_0z^{p-1} + (p+1)a_1z^p + (p+2)a_2z^{p+1} + (p+3)a_3z^{p+2} + \dots$$

$$\therefore y''(z) = p(p-1)a_0z^{p-2} + (p+1)pa_1z^{p-1} + (p+2)(p+1)a_2z^p + (p+3)(p+2)a_3z^{p+1} + \dots$$

Step 2: Substitute formulas from Step 1 into the d.e

$$zy''(z) = z\{p(p-1)a_0z^{p-2} + (p+1)pa_1z^{p-1} + (p+2)(p+1)a_2z^p + (p+3)(p+2)a_3z^{p+1} + \dots\}$$

$$2y'(z) = 2\{pa_0z^{p-1} + (p+1)a_1z^p + (p+2)a_2z^{p+1} + (p+3)a_3z^{p+2} + \dots\}$$

$$-zy(z) = -z\{a_0z^p + a_1z^{p+1} + a_2z^{p+2} + a_3z^{p+3} + \dots\}$$

$$zy''(z) = p(p-1)a_0z^{p-1} + (p+1)pa_1z^p + (p+2)(p+1)a_2z^{p+1} + (p+3)(p+2)a_3z^{p+2} + \dots$$

$$2y'(z) = 2pa_0z^{p-1} + 2(p+1)a_1z^p + 2(p+2)a_2z^{p+1} + 2(p+3)a_3z^{p+2} + \dots$$

$$-zy(z) = -a_0z^{p+1} - a_1z^{p+2} - a_2z^{p+3} - a_3z^{p+4} + \dots$$

$$\begin{array}{ccccccccccc} zy''(z) = & p(p-1)a_0z^{p-1} & + & (p+1)pa_1z^p & + & (p+2)(p+1)a_2z^{p+1} & + & (p+3)(p+2)a_3z^{p+2} & + & \dots \\ 2y'(z) = & 2pa_0z^{p-1} & + & 2(p+1)a_1z^p & + & 2(p+2)a_2z^{p+1} & + & 2(p+3)a_3z^{p+2} & + & \dots \\ -zy(z) = & & & & & -a_0z^{p+1} & & -a_1z^{p+2} & -a_2z^{p+3} & -a_3z^{p+4} & + \dots \end{array}$$

Step 3: Equate coefficients of powers of z

$$z^{p-1}: p(p-1)a_0 + 2pa_0 = 0$$

$$z^p: (p+1)pa_1 + 2(p+1)a_1 = 0$$

$$z^{p+1}: (p+2)(p+1)a_2 + 2(p+2)a_2 - a_0 = 0$$

$$z^{p+2}: (p+3)(p+2)a_3 + 2(p+3)a_3 - a_1 = 0$$

$$z^{p+3}: (p+4)(p+3)a_4 + 2(p+4)a_4 - a_2 = 0$$

$$z^{p+4}: (p+5)(p+4)a_5 + 2(p+5)a_5 - a_3 = 0$$

Step 4: Find the possible values of p by setting the lowest power of z to zero

$$p(p-1)a_0 + 2pa_0 = 0$$

$$(p^2 - p + 2p)a_0 = 0$$

$$\text{with } a_0 \neq 0$$

$$p^2 - p + 2p = 0$$

$$p^2 + p = 0$$

$$p(p+1) = 0$$

$$p = 0 \quad \text{or} \quad p + 1 = 0$$

$$p = -1$$

$\therefore p = 0$ and $p = -1$ are the solutions

Therefore, the appropriate p values in the Frobenius Form are 0 and -1 .

The difference of the two p solutions is an integer (whole number). When the p -values are an integer apart, it is likely you will only get one solution from the Frobenius Method.

As $p_1 - p_2$ is a positive integer ($0 - (-1) = 1$) there is a second, linearly independent solution of the form,

$$y_2(z) = ky_1(z) \ln(z) + z^{p_2} \sum_{n=0}^{\infty} b_n z^n$$

- (e) Finding the series solution of the differential equation near $x = -2$, where at least the first four non-zero terms are required to be included in the solution.

Step 5: Find the y values for the top (largest) p value ($p = 0$)

$$z^p: (0+1)(0)a_1 + 2(0+1)a_1 = 0$$

$$(1)(0)a_1 + 2(1)a_1 = 0$$

$$2a_1 = 0$$

$$a_1 = 0$$

$$z^{p+1}: (0+2)(0+1)a_2 + 2(0+2)a_2 - a_0 = 0$$

$$(2)(1)a_2 + 2(2)a_2 - a_0 = 0$$

$$2a_2 + 4a_2 - a_0 = 0$$

$$6a_2 - a_0 = 0$$

$$a_2 = \frac{a_0}{6}$$

$$z^{p+2}: (0+3)(0+2)a_3 + 2(0+3)a_3 - a_1 = 0$$

$$(3)(2)a_3 + 2(3)a_3 - a_1 = 0$$

$$6a_3 + 6a_3 - a_1 = 0$$

$$12a_3 - a_1 = 0$$

$$a_3 = \frac{a_1}{12}$$

$$a_3 = \frac{0}{12}$$

$$a_3 = 0$$

$$z^{p+3}: (0+4)(0+3)a_4 + 2(0+4)a_4 - a_2 = 0$$

$$(4)(3)a_4 + 2(4)a_4 - a_2 = 0$$

$$12a_4 + 8a_4 - a_2 = 0$$

$$20a_4 - a_2 = 0$$

$$a_4 = \frac{a_2}{20}$$

$$a_4 = \frac{\frac{a_0}{6}}{20}$$

$$a_4 = \frac{a_0}{120}$$

$$z^{p+4}: (0+5)(0+4)a_5 + 2(0+5)a_5 - a_3 = 0$$

$$(5)(4)a_5 + 2(5)a_5 - a_3 = 0$$

$$20a_5 + 10a_5 - a_3 = 0$$

$$30a_5 - a_3 = 0$$

$$a_5 = \frac{a_3}{30}$$

$$a_5 = \frac{0}{30}$$

$$a_5 = 0$$

$$z^{p+5}: (p+6)(p+5)a_6 + 2(p+6)a_6 - a_4 = 0$$

$$z^{p+5}: (0+6)(0+5)a_6 + 2(0+6)a_6 - a_4 = 0$$

$$(6)(5)a_6 + 2(6)a_6 - a_4 = 0$$

$$30a_6 + 12a_6 - a_4 = 0$$

$$42a_6 - a_4 = 0$$

$$a_6 = \frac{a_4}{42}$$

$$a_6 = \frac{\frac{a_0}{120}}{42}$$

$$a_6 = \frac{a_0}{5040}$$

$$\therefore y_1(z) = z^0 \left(a_0 + 0z + \frac{a_0}{6}z^2 + 0z^3 + \frac{a_0}{120}z^4 + 0z^5 + \frac{a_0}{5040}z^6 + \dots \right)$$

$$y_1(z) = a_0 + \frac{a_0}{6}z^2 + \frac{a_0}{120}z^4 + \frac{a_0}{5040}z^6 + \dots$$

$$y_1(z) = a_0 \left(1 + \frac{z^2}{3!} + \frac{z^4}{5!} + \frac{z^6}{7!} + \dots \right)$$

Transforming back to x

$$y_1(x) = a_0 \left\{ 1 + \frac{(x+2)^2}{3!} + \frac{(x+2)^4}{5!} + \frac{(x+2)^6}{7!} + \dots \right\}$$

Step 6: Try find the y values for the bottom (smallest) p value ($p = -1$)

$$z^p: (-1+1)(-1)a_1 + 2(-1+1)a_1 = 0$$

$$(0)(-1)a_1 + 2(0)a_1 = 0$$

$$0a_1 = 0$$

$\therefore a_1$ is undetermined, it can be any value

$$z^{p+1}: (-1+2)(-1+1)a_2 + 2(-1+2)a_2 - a_0 = 0$$

$$(1)(0)a_2 + 2(1)a_2 - a_0 = 0$$

$$2a_2 - a_0 = 0$$

$$a_2 = \frac{a_0}{2}$$

$$z^{p+2}: (-1+3)(-1+2)a_3 + 2(-1+3)a_3 - a_1 = 0$$

$$(2)(1)a_3 + 2(2)a_3 - a_1 = 0$$

$$2a_3 + 4a_3 - a_1 = 0$$

$$6a_3 - a_1 = 0$$

$$a_3 = \frac{a_1}{6}$$

$$y_2(z) = z^{-1} \left(a_0 + a_1 z + \frac{a_0}{2} z^2 + \frac{a_1}{6} z^3 + \dots \right)$$

$$y_2(z) = \frac{a_0}{z} + a_1 + \frac{a_0}{2} + \frac{a_1}{6} z^2 + \dots$$

$$y_2(z) = a_0 \left(\frac{1}{2} + \frac{1}{z} \right) + a_1 \left(1 + \frac{z^2}{6} \right) + \dots$$

Transforming back to x

$$y_2(x) = a_0 \left\{ \frac{1}{2} + \frac{1}{(x+2)} \right\} + a_1 \left\{ 1 + \frac{(x+2)^2}{6} \right\} + \dots$$

This is the rare case that the solutions for $p = 0$ and $p = -1$ are different despite the fact that the two p solutions differ by an integer.